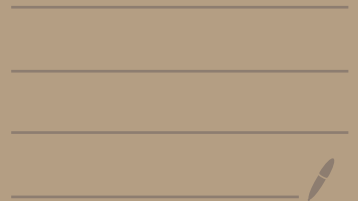


## Lec 7. Matrix Exponential.

Sec 6.4. [Smole].

Sec 6.5 Smole. Non-homogeneous.

Sec 3.1 [Teschl]



Q1: Motivation and def.

• Motivation:

1)

$$\frac{d}{dt} x(t) = a x(t) \Rightarrow x(t) = e^{at} x_0.$$

• Can we generalize to matrix?

$$\frac{d}{dt} x(t) = A x(t), \quad A \in \mathbb{R}^{n \times n}.$$

Formally, for

$$x(t) = \exp(tA) x_0$$

[Can we make sense of it?]

• Def Matrix Exponential:

$$\exp[A] = \sum_{j=0}^{\infty} \frac{1}{j!} A^j$$

①

• Well-defined: [Treatment earlier than the book].

• Convergence of ①:

$$\text{Suppose } \max |A_{ij}| \leq a.$$

Then

$$|(A^2)_{ij}| \leq \sum_{l=1}^n |A_{il} A_{lj}| \leq n \cdot a^2$$

$$\Rightarrow |(A^{k+1})_{ij}| \leq \sum_{l=1}^n |A_{il} (A^k)_{lj}| \leq |a \cdot n| \cdot (na)^{k-1} \cdot a$$

$$\stackrel{\wedge}{\text{if}} |(A^k)_{ij}| \leq (na)^{k-1} \cdot a \leq (na)^k \cdot a.$$

$\Rightarrow$  Converge.

• In general,  $\exp[A+B] \neq \exp[A] \cdot \exp[B]$ !

• Compute a few cases.

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}. \quad A = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$$

①  $e^A = \begin{pmatrix} e^\lambda & \\ & e^\mu \end{pmatrix}$

② .

$$A = \beta \cdot J, \quad J = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$$

$$J^2 = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \cdot \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} = \begin{pmatrix} -1 & \\ & -1 \end{pmatrix} = -I.$$

$$J^3 = J^2 \cdot J = -J.$$

$$J^4 = J^2 \cdot J^2 = Id.$$

thus.  $A^{4k} = I \cdot \beta^{4k}$

$$A^{4k+1} = \beta^{4k+1} \cdot J,$$

$$A^{4k+2} = -\beta^{4k+2} \cdot I$$

$$A^{4k+3} = -\beta^{4k+3} \cdot J$$

$$\exp(A) = \begin{pmatrix} \sum_{k=0}^{\infty} (-1)^k \frac{\beta^{2k}}{(2k)!} \\ - \sum_{k=0}^{\infty} (-1)^k \frac{\beta^{2k+1}}{(2k+1)!} \end{pmatrix}$$

$$\sum_{k=0}^{\infty} (-1)^k \frac{\beta^{2k+1}}{(2k+1)!}$$

$$\sum_{k=0}^{\infty} (-1)^k \frac{\beta^{2k}}{(2k)!}$$

$$= \begin{pmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{pmatrix}$$

rotation.

## II: Properties.

• Lemma 3.1. If  $AB = BA$ . Then

$$(1) B = T^{-1}AT, \quad \exp[B] = T^{-1} \exp[A] T$$

$$(2) \exp[A+B] = \exp[A] \exp[B]$$

$$(3) \exp[-A] = (\exp[A])^{-1} \rightarrow \text{Matrix inverse.}$$

Pf: (1) class

$$(2) \Rightarrow (3) \quad A \cdot (-A) = (-A) \cdot A$$

$$\exp[-A] \cdot \exp[A] = \exp[-A+A] = \exp[0] = \text{Id.}$$

Prop. If  $Av = \lambda v$   
then  $e^{Av} = e^{\lambda} v$ .

Prop.  $\frac{d}{dt} \exp[tA] = A \cdot \exp[tA] = \exp[tA] \cdot A$

$$\text{Show } \lim_{h \rightarrow 0} \frac{\exp[(t+h)A] - \exp[tA]}{h} = \exp[tA] A$$

Thm. Let  $A$  be an  $n \times n$  matrix. Then the solution to  
$$\frac{d}{dt} X = A \cdot X$$

with  $X(0) = x_0$  is  $X(t) = e^{At} \cdot x_0$ . It is unique.

$e^{At}$  is called Fundamental matrix [in BDM].

• Jordan form: (How to compute  $e^{At}$  in General)

(See IT). Section 3)

Thm (Jordan canonical form). Let  $A$  be a complex  $n \times n$  matrix, [Thus it also covers  $n \times n$  real matrix]

there exists  $U$  s.t.  $\rightarrow J$   
 $U^{-1} A U = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_m \end{pmatrix}, A = U J U^{-1}$

with each block of the form

$$J_i = \alpha I + N = \begin{pmatrix} \alpha & 1 & & \\ & \alpha & \ddots & \\ & & \ddots & 1 \\ & & & \alpha \end{pmatrix}, U = [U_1, U_2, \dots, U_n]$$

and  $\alpha$  is eigenvalue of  $A$ ,  $U_j$  generalised eigenvectors of  $A$   
( $\alpha$  can be complex) i.e.  $(A - \lambda I)^k U_j = 0$

• Now compute

$$\exp[A] = \exp[U J U^{-1}] = U \exp[J] U^{-1}$$

Remark: We do not require deriving Jordan form from any given matrix  $A$ .

• Compute  $e^{(\alpha I + N)t}$ ,  $N \in \mathbb{R}^{n \times n}$ .

• observe  $N = \begin{pmatrix} 1 & \dots & 0 \\ & \ddots & \\ & & 1 \end{pmatrix}$ .

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & \dots \\ & \ddots & & \\ & & & \end{pmatrix} \dots N^{m-1} = \begin{pmatrix} 0 & \dots & 1 \\ & 0 & \vdots \\ & & \ddots \end{pmatrix}$$

$$N^m = 0$$

$$\begin{aligned} e^{(\alpha I + N)t} &= e^{\alpha t} \left[ \sum_{i=0}^m \frac{(Nt)^i}{i!} \right] \\ &= e^{\alpha t} \begin{pmatrix} 1 & t & \frac{t^2}{2} & \dots & \frac{t^{m-1}}{(m-1)!} \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 1 \end{pmatrix} \end{aligned}$$

- Note. If  $X$  solves  $\frac{d}{dt} X = A \cdot X$ ,  $X \in \mathbb{R}^n$ .

$$\frac{d}{dt} X = A \cdot X,$$

Then

$$\begin{cases} \frac{d}{dt} \operatorname{Re} X = A \cdot \operatorname{Re} X(t) \\ \frac{d}{dt} \operatorname{Im} X = A \cdot \operatorname{Im} X(t). \end{cases}$$

If  $X(t) = u(t) + i v(t)$ . Then  $u(t), v(t)$  are real solutions to  $A$ .

- Real Jordan Block. [optional] Not required.
- If  $A$  contains complex eigenvalue  $\alpha$ , and eigenvector  $u$ .
- there is a conjugate pair  $\bar{\alpha}, \bar{u}$
- Replace  $u, \bar{u}$  in  $\mathcal{U}$  by  $\operatorname{Re}(u), \operatorname{Im}(u)$ .

one can get real Jordan block

$$A = V J V^{-1}$$

$J$  has the form

$$\alpha = a + ib$$

$$\begin{pmatrix} J_i & \\ & J_i \end{pmatrix} \xrightarrow{U} \begin{pmatrix} R & I \\ & R & \ddots \\ & & \ddots & I \\ & & & R \end{pmatrix}, \quad R = \begin{pmatrix} a & b \\ -b & a \end{pmatrix},$$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{12}$$

Back to repeated root. Consider

$$\frac{d}{dt} X = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \cdot X = AX$$

$$X(t) = e^{At} X_0 = \begin{pmatrix} e^{\lambda t} & e^{\lambda t} t \\ 0 & e^{\lambda t} \end{pmatrix} \begin{pmatrix} X_{01} \\ X_{02} \end{pmatrix}$$

get the same result.

# • Nonhomogeneous Linear eqn. / Smale. Sec 6.5

• Goal:  $X' = AX + G(t)$ .  $X \in \mathbb{R}^{n \times 1}$   
 $A \in \mathbb{R}^{n \times n}$

•  $G$  is forcing term.  $G \in \mathbb{R}^{n \times 1}$

• E.g. 2<sup>nd</sup> order linear eqn

$$y'' + py' + qy = f(t).$$

$$\begin{cases} x_1 = y \\ x_2 = y' \end{cases} \Rightarrow \frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{pmatrix} x_2 \\ -px_2 - qx_1 + f(t) \end{pmatrix}$$
$$= \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ f(t) \end{bmatrix}$$

• Variation of parameters. Consider

$$\frac{d}{dt} X(t) = A X(t) + G(t), \quad A \in \mathbb{R}^{n \times n}.$$

•  $X(0) = X_0$ . Then

$$X(t) = \exp(tA) \cdot \left[ X_0 + \int_0^t \exp(-sA) G(s) ds \right].$$

• pf: See class.

RMK: Generalize to PDG settings.